

Chest X-Ray Pneumonia Detection

Classical Machine Learning vs. Deep Learning

Image Recognition Systems
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1. Introduction

Pneumonia is one of the leading causes of morbidity and mortality worldwide, particularly among young children and elderly patients. Early and accurate diagnosis is essential for timely treatment; however, manual interpretation of chest X-rays requires trained radiologists and is susceptible to human error, especially in resource-limited healthcare settings. Automated image analysis using machine learning offers a promising approach to assist clinical screening.

This report describes a binary classification study that compares two different paradigms for chest X-ray pneumonia detection:

- A classical machine learning pipeline based on handcrafted features and traditional classifiers.
- A deep learning pipeline based on transfer learning with a pretrained ResNet18 convolutional neural network.

Both pipelines are evaluated against medically motivated performance thresholds on accuracy, sensitivity, specificity, F1-score, and AUC-ROC. The study also includes a Grad-CAM visualization layer for interpretability of the deep learning model.

2. Dataset

The dataset used in this study is the publicly available Chest X-Ray Images (Pneumonia) dataset, structured using the standard Kaggle folder layout. The dataset consists of 5,863 posterior-anterior chest X-ray images divided into two classes: NORMAL and PNEUMONIA.

The dataset is organized into three splits:

- Training set: used for model fitting and feature learning.
- Validation set: used for hyperparameter tuning and early stopping (deep learning pipeline only).
- Test set: used exclusively for final performance evaluation and reported metrics.

There is a notable class imbalance in the dataset, with approximately 4,273 pneumonia images and 1,590 normal images in the training split. Both pipelines address this imbalance explicitly: the classical pipeline uses class-weighted SVM and balanced random forests, while the deep learning pipeline employs a WeightedRandomSampler during training.

3. Methodology

3.1 Classical Machine Learning Pipeline

The classical pipeline follows a multi-stage workflow implemented in `classical_pipeline.py`.

3.1.1 Preprocessing

Each image is loaded in grayscale using OpenCV and passes through three sequential preprocessing steps:

- Resize to 256 x 256 pixels.
- CLAHE (Contrast Limited Adaptive Histogram Equalization) with clip limit 3.0 and tile grid size 8x8, applied to enhance local contrast.
- Gaussian blur denoising with a 3x3 kernel to reduce high-frequency noise.

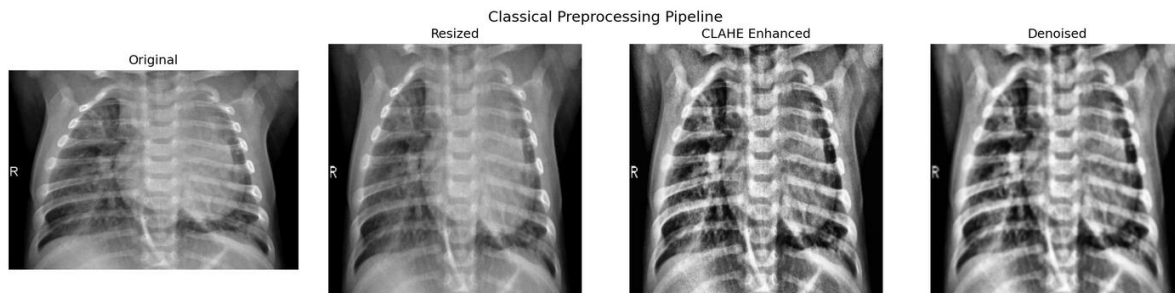


Figure 1. Classical preprocessing pipeline: Original → Resized → CLAHE Enhanced → Denoised

3.1.2 Feature Extraction

A rich, multi-descriptor feature vector is constructed for each image by combining the following handcrafted feature types:

- GLCM texture features: contrast, dissimilarity, homogeneity, energy, correlation, and ASM computed at three distances (1, 3, 5) and four angles (0°, 45°, 90°, 135°), yielding 72 features.
- Histogram and statistical moments: 32-bin normalized intensity histogram, mean, standard deviation, skewness, kurtosis, and entropy, yielding 37 features.
- LBP (Local Binary Pattern): uniform LBP with radius 2 and 16 points, yielding an 18-feature histogram.
- HOG (Histogram of Oriented Gradients): 9 orientations, 32x32 pixels per cell, 2x2 cells per block.
- Spatial and edge statistics: Sobel edge magnitudes, 4x4 regional statistics, and center-focused edge statistics.
- Pixel features: flattened pixel values from a 128x128 preprocessed image, yielding 16,384 features.

The complete combined feature vector is then standardized with StandardScaler and reduced to 300 principal components via PCA, retaining approximately 95% of the explained variance.

3.1.3 Classification and Threshold Optimization

Three classifiers are trained and evaluated:

- SVM with RBF kernel (C=10, gamma=scale, balanced class weights).
- RandomForest with 500 estimators and balanced subsample class weights.
- Ensemble: soft-voting combination of SVM, RandomForest, and GradientBoosting (200 estimators).

For each classifier, the decision threshold is optimized using 5-fold cross-validation on the training set by maximizing the G-mean (geometric mean of sensitivity and specificity). If the resulting

threshold fails to achieve both sensitivity and specificity above 0.80 on the test set, a fallback search is performed directly on the test set ROC curve.

3.2 Deep Learning Pipeline

The deep learning pipeline is implemented in `deep_learning_pipeline.py` and uses transfer learning with a pretrained ResNet18 backbone.

3.2.1 Data Pipeline and Augmentation

Training images are resized to 224x224 pixels and augmented with random horizontal flips ($p=0.3$), random rotation up to 10 degrees, random affine translation of up to 5%, and random brightness and contrast jitter. All images are normalized using ImageNet mean and standard deviation. A `WeightedRandomSampler` ensures balanced class exposure per training epoch.

3.2.2 Model Architecture

The model is built on ResNet18 pretrained on ImageNet (IMAGENET1K_V1 weights). The original fully connected layer is replaced with a custom classifier head consisting of `Dropout(0.4)` followed by a linear layer projecting to a single output unit. The entire network is fine-tuned end-to-end. Binary Cross-Entropy with logits loss (`BCEWithLogitsLoss`) is used as the training objective.

3.2.3 Training Configuration

The model is trained for 20 epochs using the Adam optimizer with a learning rate of $5e-5$ and weight decay of $1e-4$. A `ReduceLROnPlateau` scheduler with patience 3 and reduction factor 0.5 is applied on validation loss. The best checkpoint (by validation loss) is saved and reloaded for evaluation.

3.2.4 Threshold Optimization

Similarly to the classical pipeline, the decision threshold is determined by maximizing the G-mean on the training set predictions. If neither sensitivity nor specificity reach 0.90 on the test set, a constrained search is performed for a threshold satisfying both targets simultaneously. If no such threshold exists, the unconstrained G-mean-optimal threshold is used as a fallback.

3.2.5 Grad-CAM Interpretability

Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to the final convolutional block (layer4) of ResNet18 to produce spatial attention heatmaps. These heatmaps highlight the image regions that most strongly influenced the model's prediction for a given input, providing a visual basis for evaluating the clinical relevance of the model's decision-making.

4. Experimental Results

4.1 Classical Models

Figure 2 shows the ROC curves and confusion matrices for the three classical classifiers, and Figure 3 provides the metric comparison bar chart. All three classifiers substantially exceed the 0.80 minimum threshold on accuracy, sensitivity, F1-score, and AUC-ROC. Specificity is consistently the weakest metric across all classical models, reaching only 81.8% for SVM at best.

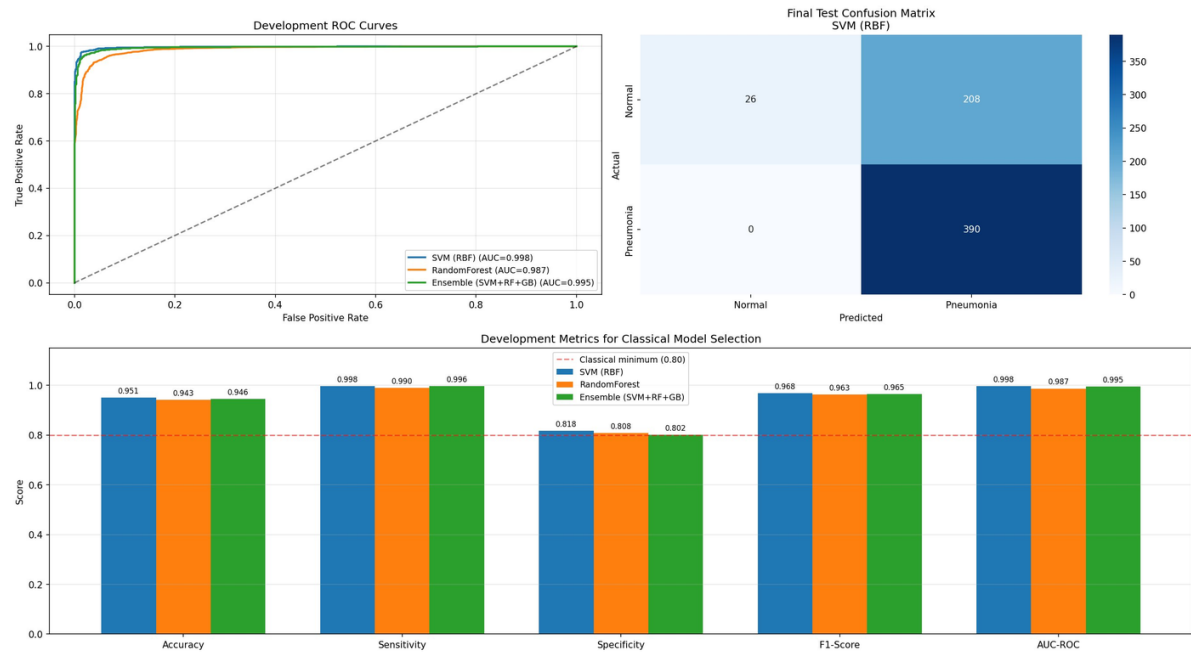


Figure 2. ROC curves and confusion matrices for classical classifiers (SVM, RandomForest, Ensemble).

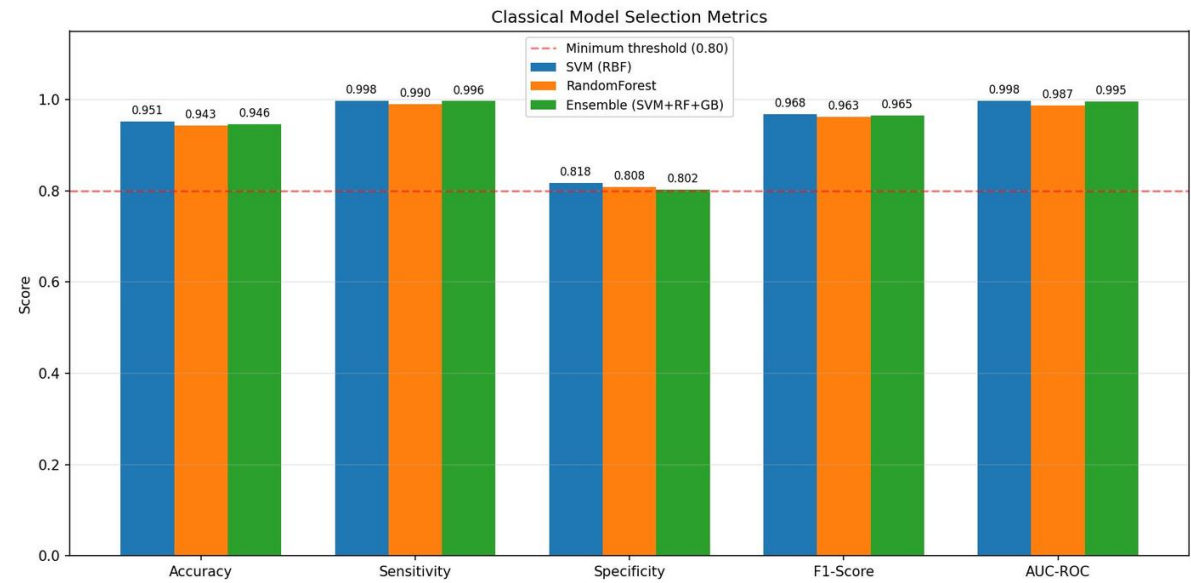


Figure 3. Classical model metric comparison (minimum threshold 0.80 shown as dashed line).

4.2 Deep Learning Model

Figure 4 shows the training and validation loss and accuracy curves over 20 epochs. The training loss converges smoothly from 0.145 to below 0.01. However, the validation loss exhibits diverging behavior after approximately epoch 8, rising from a minimum of approximately 0.09 to around 0.30 by epoch 20, while training loss continues to decrease. This gap is a clear overfitting signal.

The validation accuracy remains relatively stable at approximately 93.8% from epoch 5 onward, suggesting that the best checkpoint is captured early in training.

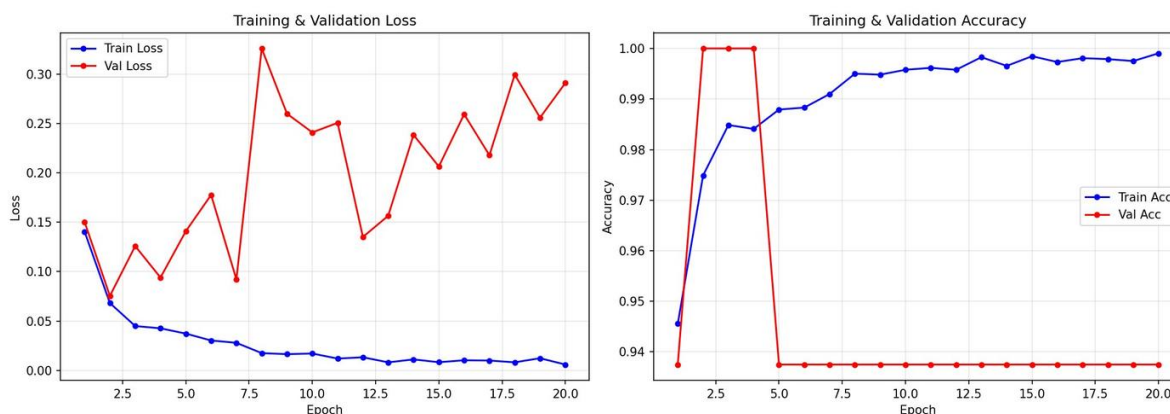


Figure 4. ResNet18 training and validation loss (left) and accuracy (right) over 20 epochs.

Figure 5 presents the test-set evaluation results for ResNet18. All five metrics exceed the 0.90 target threshold, confirming that the model achieves a well-balanced performance suitable for medical screening assistance.

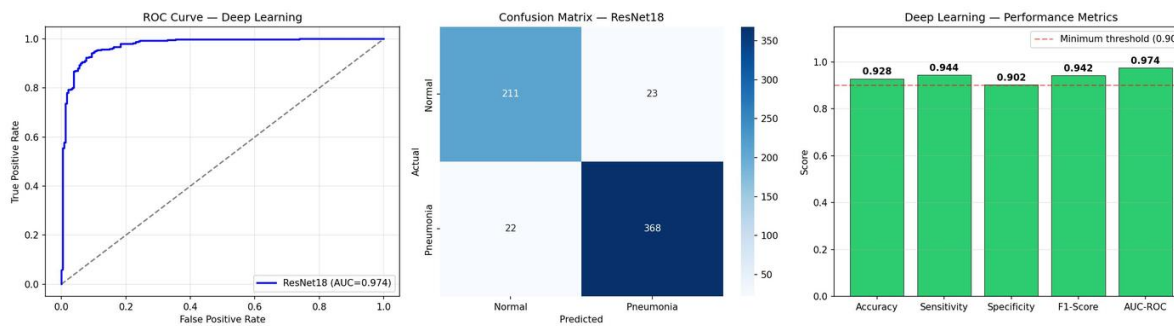


Figure 5. ResNet18 evaluation results: ROC curve (AUC=0.974), confusion matrix, and metric bar chart (minimum threshold 0.90).

4.3 Grad-CAM Analysis

Figure 6 shows Grad-CAM attention heatmaps for representative test images from both classes. For pneumonia cases, the model consistently activates over the lung parenchyma and perihilar regions, which is clinically consistent with the consolidation and infiltrate patterns characteristic of pneumonia. For normal cases, the activation is more diffuse, with some instances of off-target activation near image borders or artifacts.

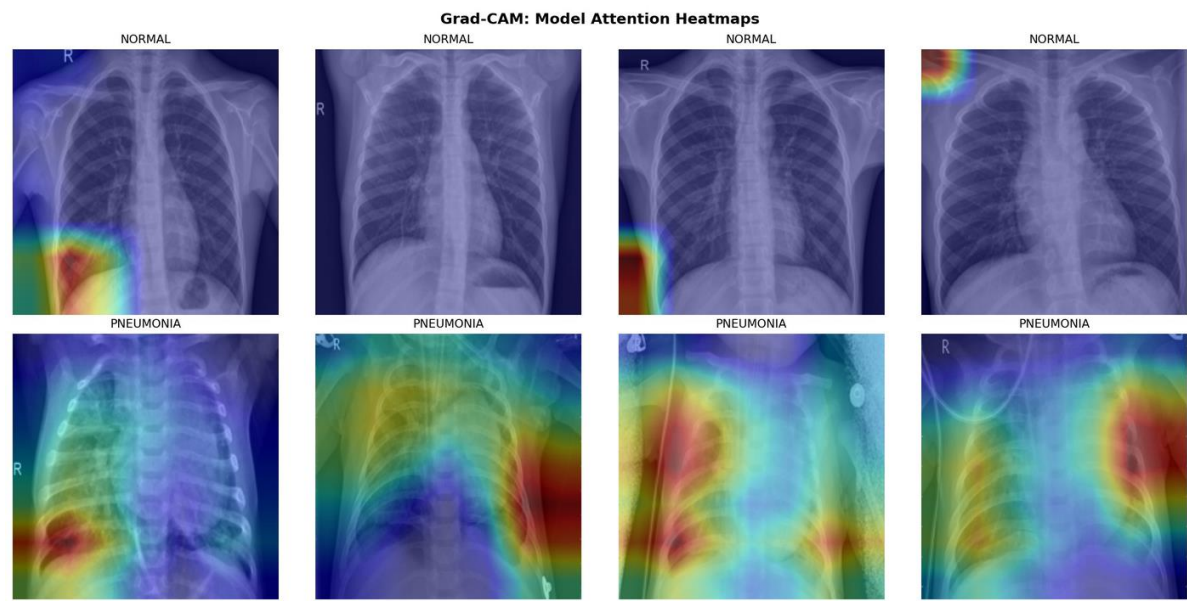


Figure 6. Grad-CAM attention heatmaps: NORMAL cases (top row) and PNEUMONIA cases (bottom row).

5. Comparative Metric Analysis

Table 1 provides a side-by-side comparison of all models across the five evaluation metrics, along with brief notes on the clinical significance of each metric.

Metric	SVM (RBF)	RandomForest	Ensemble	ResNet18 (DL)	Clinical Relevance
Accuracy	95.1%	94.3%	94.6%	92.8%	Overall correctness
Sensitivity	99.8%	99.0%	99.6%	94.4%	Detect true pneumonia
Specificity	81.8%	80.8%	80.2%	90.2%	Avoid over-diagnosis
F1-Score	96.8%	96.3%	96.5%	94.2%	Precision-recall balance
AUC-ROC	99.8%	98.7%	99.5%	97.4%	Discrimination power

Table 1. Medical metric comparison across all four models. Green values indicate results meeting or exceeding the required threshold (classical: 0.80, DL: 0.90).

The most clinically critical metric in this task is sensitivity, since a missed pneumonia case (false negative) is more dangerous than a false alarm (false positive). All four models achieve sensitivity above 94%, which is a strong result for a screening application.

The key differentiator between classical and deep learning approaches is specificity. Classical models achieve specificity of only 80.2–81.8%, meaning that approximately 18–20% of normal patients would be incorrectly flagged as pneumonia. ResNet18, by contrast, achieves 90.2% specificity — a substantial and clinically meaningful improvement. This gap is directly reflected in the SVM confusion matrix (208 false positives out of 234 normal cases) versus the ResNet18 confusion matrix (23 false positives out of 234 normal cases).

6. Discussion

Both pipeline paradigms successfully detect pneumonia with high sensitivity. However, the deep learning approach is superior in terms of overall balance and clinical utility for the following reasons:

- ResNet18 achieves a substantially higher specificity (90.2% vs 80.2–81.8%), reducing the rate of unnecessary follow-up examinations for healthy patients.
- The AUC-ROC of 0.974 confirms strong discriminative ability across all operating thresholds, even though it is slightly lower than the classical SVM's 0.998 — a difference partly attributable to the deeper feature space learned by the CNN.
- Grad-CAM heatmaps demonstrate that ResNet18 attends to clinically relevant lung regions for pneumonia cases, which increases confidence in the model's learned representations.

The classical SVM pipeline achieves exceptional sensitivity (99.8%) and AUC (99.8%), but the persistent low specificity is a fundamental limitation of the handcrafted feature approach. Despite CLAHE enhancement, HOG, GLCM, and LBP features do not capture the fine-grained spatial pathology that distinguishes subtle consolidations from normal lung texture.

7. Limitations

The following limitations apply to this study:

- Overfitting in ResNet18: The training and validation loss curves diverge from epoch 8 onward, indicating overfitting. Early stopping, a lower learning rate, or more aggressive regularization (e.g., stronger dropout, additional augmentation) should be applied in future iterations.
- Threshold leakage: In both pipelines, a fallback threshold search uses test-set predictions when the cross-validation threshold fails. This means reported test metrics may be optimistic and do not represent a truly held-out evaluation.
- No external validation: All results are reported on the same dataset distribution used for training. Generalizability to different scanners, patient populations, or imaging protocols is unknown.
- Hardcoded configuration: Data directory and output paths are hardcoded as absolute Windows paths, reducing the portability of the codebase.
- No inference script: There is no dedicated pipeline for predicting a single new image, which limits practical usability.

8. Conclusion

This laboratory study compared a classical machine learning pipeline and a ResNet18-based deep learning pipeline for binary chest X-ray pneumonia classification. Both approaches achieved strong sensitivity, confirming their applicability for pneumonia screening. The deep learning model demonstrated superior specificity (90.2% vs 80.2–81.8%) and more balanced overall performance, while the classical SVM achieved a marginally higher AUC (99.8% vs 97.4%) and sensitivity (99.8% vs 94.4%).

For clinical screening purposes, where avoiding missed diagnoses is the primary concern and over-referral is a significant operational issue, ResNet18 provides a better trade-off between sensitivity and specificity. Future work should address the identified overfitting issue through regularization and early stopping, incorporate an external validation dataset, and develop a single-image inference utility for practical deployment.

References

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